

# DIMAS DEFFIEUX

Undergrad Student interested in Multilingual and Low-Resource NLP looking for an internship in ML Research or SWE  
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LinkedIn | Github | Portfolio

## EDUCATION

**CentraleSupélec & McGill University**  
*Gif-sur-Yvette, France & Montréal, Canada*

Ongoing

- Bachelor of Global Engineering. GPA: 3.33
- Relevant coursework: Algorithms, Differential Calculus, Probability, Group Programming Competition, Linear Algebra.

**The Village School**  
*Houston, Texas, USA*

2021–2025

- IB Diploma. Computer Science Honors Society (2× Best Programmer award).
- Founder of Coding Club; VP of Table Tennis Club.

## PROJECTS & EXPERIENCE

**OCR Evaluation Benchmark**  
*Ongoing*

Sep 2026 – Present

- Designing a stratified evaluation set of 1600 images across website, video game, and video screenshots, real life signage, and stylized text, with hallucination-trap and stress-case subsets.
- Benchmarking PP-OCrv5, frontier VLM's, and PaddleOCR-VL on CER and chrF++, along with cost per 1,000 pages and throughput.

**SLB (Schlumberger) – Software Engineering Intern**

Jun 2026 – Jul 2026

- Designed an agentic chatbot system to search efficiently through internal SharePoint Documents
- Practiced system design principles to provide effective results catered to requirements given through interviews with managers
- Implemented system with Microsoft Copilot Studio to improve modularity and maintainability for the less technical department after my departure, in addition for integration with already in place M365 architecture
- Improved AI fluency among the department by providing a simple prompt engineering guide, and consulting some workers on how to automate and leverage AI

**Bureau des Bachelors – Web Infrastructure Overhaul**  
*Ongoing*

Feb 2026 – Present

- Conducting a full-stack overhaul focused on sophisticated, maintainable, well-documented code, replacing a disorganized legacy codebase to enable smooth handoffs to incoming developers.
- Streamlining web infrastructure by migrating from a fragmented PHP setup to a modular Next.js backend, reducing separately hosted websites from 4 to 2.
- Redesigning website and functionality, integrating event signups, member subscription management, and admin functionalities.
- Enhancing database efficiency by optimizing dataflow and sunseting unused features.

**English-French Transformer Translation Model**

Oct 2025 – Feb 2026

- Built a translation model from scratch using the Transformer architecture in PyTorch, with HuggingFace tokenizers for data preprocessing.
- Implemented multi-head self-attention, positional encoding, and a full encoder-decoder architecture with greedy and beam search decoding.
- Developed the complete end-to-end pipeline: data preprocessing, architecture definition, training loop, and evaluation.

**Community Forum Web App**  
*Course Project*

Mar 2023 – Apr 2023

- Built a full-stack community forum in Java using Jakarta Servlet, MySQL, and GlassFish, implementing MVC architecture.
- Implemented basic CRUD functions for managing posts and comments on the forum

## EXTRA-CURRICULAR INVOLVEMENT

**Automatants – Technical Member**  
*Pole Geek*

Oct 2025 – Present

- Technical member of CentraleSupélec's machine learning student association.

**ViaRezo – Technical Developer**  
*Pole Dev*

Sept 2025 – Present

- Developer for the student-run ISP and infrastructure association.
- Attended trainings to improve knowledge in infrastructure (Datadog overview, Firewalls)

**Bureau des Bachelors – VP of IT**

Feb 2026 – Present

- Leading IT and web infrastructure for the Bachelor student office.

## SKILLS

**ML/AI:** PyTorch, HuggingFace, RAG, Ollama, TensorFlow, Keras, scikit-learn, matplotlib, pandas, NumPy.

**Web:** Java, FastAPI, Next.js, React.js, Jakarta Servlet, JavaScript, HTML/CSS, MySQL, MVC Architecture.

**Tools:** Git, Linux, Python.

**Languages:** English (Native), French (Native), Korean (Estimated TOPIK 3), Japanese (Estimated N4).

## INTERESTS

Multilingual/Low-Resource NLP, Research, AI/Machine Learning, Software development, Language acquisition, Weight training, Calisthenics, Mechanical Keyboards.